

The Sharp Logarithmic Tauberian Gap: Signed Cesàro Defects, Dyadic Countermodels, and the Literal Fixed-Shift Affine Gate

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July 2026

Abstract

Let

$$S_N = \sum_{n \leq N} a_n, \quad L_N = \sum_{n \leq N} \frac{a_n}{n}, \quad L_0 = 0.$$

We isolate the exact discrete Tauberian interface between logarithmic and ordinary cancellation. Summation by parts gives the identity

$$S_N = NL_N - \sum_{m < N} L_m, \quad \frac{S_N}{N} = L_N - \frac{1}{N} \sum_{m < N} L_m.$$

Consequently ordinary cancellation is equivalent, with identical rate, to the vanishing of the *signed backward Cesàro defect* of the unnormalized logarithmic primitive L . Convergence of L_N is sufficient by Cesàro convergence, but boundedness is not. Indeed, for

$$a_n = (-1)^k \quad (2^k \leq n < 2^{k+1}),$$

one has $|a_n| = 1$, $L_N = O(1)$, and even

$$\frac{1}{\log \omega(x)} \sum_{x/\omega(x) < n \leq x} \frac{a_n}{n} \rightarrow 0$$

for every $1 \leq \omega(x) \leq x$ with $\omega(x) \rightarrow \infty$, whereas S_N/N has subsequential limits $1/3$ and $-1/3$. Thus normalized logarithmic cancellation alone has no structure-free implication for the ordinary mean.

We then give quantitative positive criteria. The absolute Cesàro oscillation dominates $|S_N|/N$. If

$$\omega_N(L) := \max_{\lfloor N/2 \rfloor \leq m \leq N} |L_N - L_m| \rightarrow 0,$$

then $S_N/N \rightarrow 0$; a power bound $\omega_N(L) \ll N^{-\eta}$, $0 < \eta < 1$, transfers with the same power. A second criterion uses the dyadic logarithmic variation

$$V_k = \sum_{2^k \leq n < 2^{k+1}} \frac{|a_n|}{n}.$$

We prove an explicit geometrically weighted convolution bound, hence $V_k \rightarrow 0$ also suffices.

Finally, we audit what these identities do and do not supply for the literal growing fixed- h_0 affine Möbius packet. The exact Tauberian statements and countermodel are L0. Their faithful attachment to every physical fiber and the complete outer reassembly is an L1 certificate. Establishing the required uniform terminal cancellation for the actual growing affine family is an unproved L2 arithmetic problem. No parity breach, prime-pair lower bound, or twin-prime conclusion is claimed.

Keywords: logarithmic average; Tauberian theorem; Cesàro defect; dyadic oscillation; local variation; fixed shift; affine Möbius correlation.

Convention. Every physical statement retains one fixed nonzero integer shift h_0 , the literal growing affine coefficient, its actual interval and masks, all native keys, both polarizations, and one global post-aggregation normalization. A result for fixed forms, averaged shifts, almost all scales, almost all interval origins, or a structure-free sequence is not identified with the literal packet.

1 Introduction

Logarithmic averaging is often substantially more accessible than ordinary averaging in multiplicative number theory. In particular, Tao proved logarithmically averaged two-point Chowla and Elliott theorems for fixed affine forms [4]. The literal fixed-shift route considered in TPC-32 and TPC-33, however, returns ordinary sums over a growing family of affine columns, with a prescribed nonzero determinant h_0 , actual masks, short fibers, and a complete outer reassembly [6, 7].

TPC-85 exhibited a bounded dyadic sequence showing that no structure-free black box simply deletes the logarithmic weight [10]. The purpose of the present paper is to sharpen that obstruction into an exact theorem:

the missing datum is the signed Cesàro defect of the *unnormalized* logarithmic primitive.

This distinction matters. If

$$H_N = \sum_{n \leq N} \frac{1}{n}, \quad \lambda_N = \frac{1}{H_N} \sum_{n \leq N} \frac{a_n}{n},$$

then λ_N is the normalized logarithmic mean, while

$$L_N = H_N \lambda_N$$

is the unnormalized logarithmic primitive. The exact discrete Abel identity is written in terms of L_N , not λ_N .

TPC-86 develops a different positive interface: coherent Fourier reassembly of the distinguished zero mode from a controlled low-frequency window [11]. The present paper does not replace that route. It audits the separate possibility of obtaining ordinary affine cancellation from logarithmically averaged correlation information.

1.1 Main results

For an arbitrary complex sequence $a = (a_n)$, define

$$S_N = \sum_{n=1}^N a_n, \quad L_N = \sum_{n=1}^N \frac{a_n}{n}, \quad S_0 = L_0 = 0.$$

Our first result is the pointwise identity

$$\frac{S_N}{N} = L_N - \frac{1}{N} \sum_{m=0}^{N-1} L_m = \frac{1}{N} \sum_{m=0}^{N-1} (L_N - L_m). \quad (1)$$

Therefore $S_N/N \rightarrow 0$ if and only if the signed quantity on the right tends to zero. This is not merely an implication: the two objects are equal for every N , so there is no hidden exponent loss.

Convergence $L_N \rightarrow \ell$ is sufficient. Boundedness $L_N = O(1)$ is sharply insufficient. The dyadic block sequence

$$a_n = (-1)^{\lfloor \log_2 n \rfloor}$$

has a bounded logarithmic primitive and normalized logarithmic mean $\lambda_N \rightarrow 0$, yet its natural means oscillate between $\pm 1/3$ along complete dyadic blocks. More strongly, every terminal multiplicative interval has uniformly bounded logarithmic sum, so every Tao-shaped normalized window with $\omega(x) \rightarrow \infty$ has cancellation.

The exact criterion also identifies useful additional hypotheses. We prove:

- (i) an absolute Cesàro oscillation criterion;
- (ii) a terminal-half local criterion

$$\max_{N/2 \leq m \leq N} \left| \sum_{m < n \leq N} \frac{a_n}{n} \right| \rightarrow 0;$$

- (iii) a rate-preserving version for fixed powers below one;
- (iv) an explicit dyadic variation convolution bound.

These are genuine Tauberian inputs: none follows from boundedness of the global logarithmic primitive alone.

1.2 Arithmetic status

The results have three proof levels.

L0: finite discrete identities, quantitative inequalities, and structure-free countermodels;

L1: a literal transfer certificate retaining the same fixed h_0 , growing affine coefficients, actual fibers, masks, outer reassembly, and global normalization;

L2: a new uniform Möbius-correlation estimate for that actual growing family.

This paper proves L0 results and formulates the L1 certificate. It proves no L2 estimate. In particular, the dyadic sequence is not multiplicative and does not obstruct a proof that uses Möbius-specific structure.

1.3 Organization

[Section 2](#) proves the exact Abel identities. [Section 3](#) gives the necessary-and-sufficient criterion and the convergence theorem. Quantitative local and variation conditions appear in [section 4](#). [Section 5](#) proves the sharp dyadic counterexample. The theorem-quantifier audit is [section 6](#); the literal physical certificate and endpoint ledger are in [section 7](#). The claim boundary is recorded in [section 8](#).

2 The exact discrete Abel interface

No boundedness, multiplicativity, or reality assumption is required in this section.

Definition 2.1 (Three means). For $a_n \in \mathbb{C}$, put

$$S_0 = L_0 = 0, \quad S_N = \sum_{n=1}^N a_n, \quad L_N = \sum_{n=1}^N \frac{a_n}{n}.$$

Also put

$$H_N = \sum_{n=1}^N \frac{1}{n}, \quad \lambda_N = \frac{L_N}{H_N}.$$

Thus S_N/N is the ordinary mean, L_N is the unnormalized logarithmic primitive, and λ_N is the normalized logarithmic mean.

The increment of the logarithmic primitive is

$$\Delta L_n := L_n - L_{n-1} = \frac{a_n}{n}. \tag{2}$$

Theorem 2.2 (Exact discrete Abel identities). *For every $N \geq 1$,*

$$S_N = \sum_{n=1}^N n \Delta L_n = N L_N - \sum_{m=0}^{N-1} L_m, \quad (3)$$

$$\frac{S_N}{N} = L_N - \frac{1}{N} \sum_{m=0}^{N-1} L_m = \frac{1}{N} \sum_{m=0}^{N-1} (L_N - L_m). \quad (4)$$

Conversely,

$$L_N = \frac{S_N}{N} + \sum_{n=1}^{N-1} \frac{S_n}{n(n+1)}. \quad (5)$$

Proof. By (2),

$$\sum_{n=1}^N n \Delta L_n = \sum_{n=1}^N n L_n - \sum_{m=0}^{N-1} (m+1) L_m = N L_N - \sum_{m=0}^{N-1} L_m.$$

The left side is S_N , proving (3). Dividing by N and rearranging gives (4). For the reverse identity, write $a_n = S_n - S_{n-1}$ and sum by parts:

$$\sum_{n=1}^N \frac{S_n - S_{n-1}}{n} = \frac{S_N}{N} + \sum_{n=1}^{N-1} S_n \left(\frac{1}{n} - \frac{1}{n+1} \right).$$

□

Corollary 2.3 (Exact normalized formulation). *With the notation of [definition 2.1](#),*

$$\frac{S_N}{N} = H_N \lambda_N - \frac{1}{N} \sum_{m=0}^{N-1} H_m \lambda_m, \quad H_0 \lambda_0 := 0. \quad (6)$$

Proof. Substitute $L_m = H_m \lambda_m$ into (4). □

Remark 2.4 (Normalization is not cosmetic). The right side of (6) is not λ_N minus the Cesàro mean of λ . The harmonic factor H_m remains inside both terms. Therefore $\lambda_N \rightarrow 0$ alone does not imply ordinary cancellation. [Section 5](#) gives a bounded-coefficient example.

Remark 2.5 (Indexing). The backward average in (4) has denominator N and contains $L_0, \dots, L_{N-1}$, but not L_N . Replacing it by the conventional average $N^{-1} \sum_{m=1}^N L_m$ changes the identity by L_N/N , which cannot be discarded without an additional growth hypothesis.

3 The necessary-and-sufficient Cesàro oscillation criterion

Definition 3.1 (Signed defect and absolute oscillation). For a sequence $L = (L_N)_{N \geq 0}$, define

$$\mathcal{T}_N(L) := L_N - \frac{1}{N} \sum_{m=0}^{N-1} L_m, \quad (7)$$

$$\mathcal{A}_N(L) := \frac{1}{N} \sum_{m=0}^{N-1} |L_N - L_m|. \quad (8)$$

We call $\mathcal{T}_N$ the signed backward Cesàro defect and $\mathcal{A}_N$ the absolute backward Cesàro oscillation.

Theorem 3.2 (Sharp discrete Tauberian criterion). *For every complex sequence a ,*

$$\boxed{\frac{S_N}{N} = \mathcal{T}_N(L)}. \quad (9)$$

Consequently,

$$\frac{S_N}{N} \rightarrow 0 \iff \mathcal{T}_N(L) \rightarrow 0,$$

and any quantitative rate on one side is exactly the same rate on the other.

Proof. This is (4). □

Corollary 3.3 (Absolute oscillation is sufficient). *For every $N \geq 1$,*

$$\frac{|S_N|}{N} \leq \mathcal{A}_N(L). \quad (10)$$

In particular, $\mathcal{A}_N(L) \rightarrow 0$ implies $S_N/N \rightarrow 0$.

Proof. Take absolute values in the last expression in (4). □

Remark 3.4 (Why the signed adjective is essential). The necessary-and-sufficient object is $\mathcal{T}_N$, not $\mathcal{A}_N$. Cancellation among the differences $L_N - L_m$ can make $\mathcal{T}_N$ small while their absolute mean is not. Thus $\mathcal{A}_N \rightarrow 0$ is a useful stronger hypothesis, but it is not asserted to be necessary.

Corollary 3.5 (Convergent logarithmic primitive). *If $L_N \rightarrow \ell \in \mathbb{C}$, then $S_N/N \rightarrow 0$. More precisely, with $e_m = |L_m - \ell|$,*

$$\frac{|S_N|}{N} \leq e_N + \frac{1}{N} \sum_{m=1}^{N-1} e_m + \frac{|\ell|}{N}. \quad (11)$$

If

$$\varepsilon_M = \sup_{n \geq M} |L_n - \ell|,$$

then, for $N > M$,

$$\frac{|S_N|}{N} \leq 2\varepsilon_M + \frac{|\ell| + \sum_{m=1}^{M-1} |L_m - \ell|}{N}. \quad (12)$$

Proof. Subtract ℓ in both terms of (4) and use the triangle inequality. This proves (11). For (12), split the backward average at M . The two tail contributions are bounded by ε_M , while the finite prefix has the displayed cost. □

Remark 3.6 (Convergence is sufficient, not necessary). Let $a_1 = 0$ and $a_n = 1/\log n$ for $n \geq 2$. Since $a_n \rightarrow 0$, Cesàro convergence gives $S_N/N \rightarrow 0$, whereas

$$L_N = \sum_{2 \leq n \leq N} \frac{1}{n \log n} = \log \log N + O(1)$$

diverges. Thus the exact criterion is strictly weaker than convergence of L_N .

Remark 3.7 (Classical terminology). The implication in [corollary 3.5](#) is the elementary discrete Kronecker lemma. We prove it here because the exact defect and its normalization, rather than the name of the lemma, are the relevant physical interface; see also Hardy [1].

4 Quantitative local and variation hypotheses

The exact criterion $\mathcal{T}_N(L) \rightarrow 0$ can be difficult to verify directly. This section gives two stronger but local conditions with explicit costs.

4.1 Terminal-half logarithmic cancellation

Definition 4.1 (Terminal-half modulus). For $N \geq 2$, put

$$\omega_N(L) := \max_{\lfloor N/2 \rfloor \leq m \leq N} |L_N - L_m| = \max_{\lfloor N/2 \rfloor \leq m \leq N} \left| \sum_{m < n \leq N} \frac{a_n}{n} \right|. \quad (13)$$

Lemma 4.2 (One-step localization). *Let $M = \lfloor N/2 \rfloor$. Then*

$$\mathcal{A}_N(L) \leq \omega_N(L) + \frac{M}{N} \mathcal{A}_M(L). \quad (14)$$

Proof. For $m < M$,

$$|L_N - L_m| \leq |L_N - L_M| + |L_M - L_m| \leq \omega_N(L) + |L_M - L_m|.$$

For $M \leq m < N$, the first term alone is at most $\omega_N(L)$. Summing both ranges and dividing by N gives (14). $\square$

Theorem 4.3 (Geometric local-to-global transfer). *Let $N_0 = N$ and $N_{j+1} = \lfloor N_j/2 \rfloor$, stopping when $N_J = 1$. Then*

$$\frac{|S_N|}{N} \leq \mathcal{A}_N(L) \leq \sum_{j=0}^{J-1} \frac{N_j}{N} \omega_{N_j}(L) + \frac{\mathcal{A}_1(L)}{N}. \quad (15)$$

In particular,

$$\omega_N(L) \longrightarrow 0 \implies \frac{S_N}{N} \longrightarrow 0. \quad (16)$$

Proof. Iterate (14). The product of the successive floor ratios telescopes to N_j/N , giving (15). Also $N_j/N \leq 2^{-j}$.

To prove the last assertion, fix $\varepsilon > 0$ and choose R so that $\omega_m(L) \leq \varepsilon$ for $m \geq R$. The terms with $N_j \geq R$ contribute at most 2ε . The remaining terms involve only finitely many moduli below R ; their numerators are bounded independently of N , and their total coefficient is $O_R(N^{-1})$. Now let $N \rightarrow \infty$, then $\varepsilon \rightarrow 0$. $\square$

Corollary 4.4 (Fixed-power transfer). *Suppose $0 < \eta < 1$ and*

$$\omega_N(L) \leq CN^{-\eta} \quad (N \geq 2).$$

Then

$$\frac{|S_N|}{N} \leq \frac{C}{1 - 2^{-(1-\eta)}} N^{-\eta} + O_L\left(\frac{1}{N}\right). \quad (17)$$

Thus the local-to-global conversion loses no fixed power below one.

Proof. In (15), use

$$\frac{N_j}{N} \omega_{N_j}(L) \leq \frac{C}{N} N_j^{1-\eta} \leq CN^{-\eta} 2^{-j(1-\eta)}$$

and sum the geometric series. $\square$

Remark 4.5 (The missing fixed-ratio datum). A theorem on windows $(x/\Omega(x), x]$ with $\Omega(x) \rightarrow \infty$ does not automatically give $\omega_N(L) \rightarrow 0$, which is a fixed-ratio window with ratio $2 + O(N^{-1})$ after endpoint rounding, and an $o(1)$, rather than $o(\log \Omega)$, target. This is the precise local Tauberian datum exposed by [theorem 4.3](#).

4.2 Dyadic logarithmic variation

Definition 4.6 (Dyadic variation). For $k \geq 0$, define

$$V_k(L) := \sum_{2^k \leq n < 2^{k+1}} |\Delta L_n| = \sum_{2^k \leq n < 2^{k+1}} \frac{|a_n|}{n}. \quad (18)$$

Theorem 4.7 (Variation convolution bound). *If $2^K \leq N < 2^{K+1}$, then*

$$\boxed{\frac{|S_N|}{N} \leq V_K(L) + \sum_{k=0}^{K-1} 2^{k+1-K} V_k(L) = V_K(L) + 2 \sum_{j=1}^K 2^{-j} V_{K-j}(L).} \quad (19)$$

Consequently,

$$V_k(L) \longrightarrow 0 \implies \frac{S_N}{N} \longrightarrow 0. \quad (20)$$

Proof. Use $S_N = \sum_{n \leq N} n \Delta L_n$. On a completed block $2^k \leq n < 2^{k+1}$ with $k < K$,

$$\frac{n}{N} \leq 2^{k+1-K}.$$

On the terminal block $n/N \leq 1$. Taking absolute values block by block proves (19).

For the final assertion, the last expression in (19) is a geometric convolution. Split it into the finitely many recent terms, where $V_{K-j} \rightarrow 0$, and the geometrically small tail. The finitely many initial values are multiplied by $O(2^{-K})$. $\square$

Corollary 4.8 (Variation power rate). *If $0 < \eta < 1$ and $V_k(L) \leq C2^{-\eta k}$, then*

$$\frac{|S_N|}{N} \ll_{\eta, C} N^{-\eta}.$$

Proof. Insert the hypothesis into (19) and sum $\sum_{j \geq 1} 2^{-j} 2^{\eta j}$. $\square$

Remark 4.9 (Strength of the variation condition). The condition $V_k \rightarrow 0$ is an absolute-variation hypothesis and is stronger than signed terminal cancellation. It is useful because it is representation-stable and quantitative, not because it is expected automatically for Möbius signs. The dyadic counterexample below has $V_k \asymp 1$.

5 A sharp bounded dyadic counterexample

Definition 5.1 (Alternating dyadic blocks). For $n \geq 1$, define

$$a_n = (-1)^k \quad \text{when } 2^k \leq n < 2^{k+1}. \quad (21)$$

Thus $|a_n| = 1$.

Write

$$h_k := \sum_{n=2^k}^{2^{k+1}-1} \frac{1}{n}. \quad (22)$$

Integral comparison gives

$$\log 2 \leq h_k < \log 2 + 2^{-k}. \quad (23)$$

Theorem 5.2 (Bounded logarithmic primitive, oscillating natural mean). *For the sequence (21):*

- (i) $L_N = O(1)$;
- (ii) $\lambda_N = L_N/H_N \rightarrow 0$;
- (iii) for $N_K = 2^{K+1} - 1$,

$$S_{N_K} = \frac{1 - (-2)^{K+1}}{3}, \quad (24)$$

and hence

$$\frac{S_{N_K}}{N_K} \longrightarrow \begin{cases} 1/3, & K \text{ even,} \\ -1/3, & K \text{ odd.} \end{cases} \quad (25)$$

In particular, boundedness of L_N , and even convergence of the normalized logarithmic mean to zero, does not imply ordinary cancellation.

Proof. If $2^K \leq N < 2^{K+1}$, then

$$L_N = \sum_{k=0}^{K-1} (-1)^k h_k + (-1)^K \sum_{n=2^K}^N \frac{1}{n}.$$

Write $h_k = \log 2 + e_k$, where $0 \leq e_k < 2^{-k}$. The alternating sum of the constant $\log 2$ terms is bounded, the error series is absolutely summable, and the final partial block has bounded harmonic mass. This proves $L_N = O(1)$. Since $H_N \rightarrow \infty$, it also proves $\lambda_N \rightarrow 0$.

At the right endpoint of the K -th complete block,

$$S_{N_K} = \sum_{k=0}^K (-1)^k 2^k = \frac{1 - (-2)^{K+1}}{3}.$$

Dividing by $2^{K+1} - 1$ gives (25). □

Theorem 5.3 (Uniform Tao-shaped logarithmic windows). *There is an absolute constant K_0 such that, for every $x \geq 2$ and $1 \leq \Omega \leq x$,*

$$\left| \sum_{x/\Omega < n \leq x} \frac{a_n}{n} \right| \leq K_0. \quad (26)$$

Consequently, for every function $1 \leq \omega(x) \leq x$ with $\omega(x) \rightarrow \infty$,

$$\frac{1}{\log \omega(x)} \sum_{x/\omega(x) < n \leq x} \frac{a_n}{n} \rightarrow 0. \quad (27)$$

Proof. The interval $(x/\Omega, x]$ meets at most two dyadic blocks partially. Each partial block has harmonic mass at most $\log 2 + 1$. The remaining complete blocks form a consecutive range. By (23), their contribution is an alternating sum of copies of $\log 2$, of absolute value at most $\log 2$, plus an absolutely summable error of total size at most 2. Thus (26) holds, for example with $K_0 = 8$. Divide by $\log \omega(x) \rightarrow \infty$. □

Corollary 5.4 (Sharp black-box failure). *For bounded sequences in general, even normalized logarithmic cancellation on every growing terminal multiplicative window does not imply $S_N/N \rightarrow 0$.*

Proof. Combine theorems 5.2 and 5.3. □

Remark 5.5 (Why this does not contradict the positive criteria). At the end of a dyadic block, the terminal half is essentially one constant-sign block, so $\omega_N(L)$ stays bounded below by a positive constant close to $\log 2$. Also

$$V_k = \sum_{2^k \leq n < 2^{k+1}} \frac{1}{n} = h_k \rightarrow \log 2.$$

Thus the counterexample fails both additional hypotheses in section 4.

Remark 5.6 (Scope). The sequence (21) is not multiplicative. It proves a logical nonimplication for structure-free Tauberian arguments. It neither models the literal Möbius coefficient nor rules out a conversion using multiplicativity, entropy decrement, pretentious distance, or another arithmetic input.

6 Quantifier audit of available arithmetic theorems

The Tauberian identity is exact only after the same ordered coefficient sequence has been fixed. A theorem that changes the forms, shift, scale, interval origin, or outer averaging measure does not automatically control its signed Cesàro defect.

6.1 Fixed-form logarithmic correlation

For fixed $a_1, a_2 \in \mathbb{N}$ and $b_1, b_2 \in \mathbb{Z}_{\geq 0}$ with $a_1 b_2 - a_2 b_1 \neq 0$, Tao proved

$$\sum_{x/\omega(x) < n \leq x} \frac{\lambda(a_1 n + b_1) \lambda(a_2 n + b_2)}{n} = o(\log \omega(x)), \quad (28)$$

where $1 \leq \omega(x) \leq x$ and $\omega(x) \rightarrow \infty$ [4]. This is a genuine two-point cancellation theorem in its stated logarithmic setting.

Equation (28) does not by itself verify $\omega_N(L) \rightarrow 0$. Its ratio $\omega(x)$ must grow, its conclusion is normalized by $\log \omega(x)$, and the affine coefficients are fixed. The literal packet instead contains polynomially growing slopes and intercepts, fixed-ratio terminal windows, actual masks, and a quantitative outer sum. The dyadic counterexample proves that this mismatch cannot be repaired for arbitrary bounded sequences by a Tauberian black box alone.

6.2 Averaged shifts, almost-all scales, and short intervals

For fixed k and $1 \leq H = H(X) \leq X$ with $H(X) \rightarrow \infty$, Matomäki, Radziwiłł, and Tao proved

$$\sum_{h_1, \dots, h_k \leq H} \left| \sum_{n \leq X} \lambda(n + h_1) \cdots \lambda(n + h_k) \right| = o(H^k X) \quad (29)$$

[3]. This is ordinary cancellation averaged over shifts; it does not select one prescribed h_0 .

Tao and Teräväinen proved unweighted Chowla and Elliott cases at all scales outside a set of zero logarithmic density. For fixed Liouville shifts, their stated Chowla cases include odd order and order two [5]. A prescribed sparse sequence of physical scales can lie inside the exceptional set, so an additional scale-localization theorem is required.

Matomäki and Radziwiłł proved mean-value control for multiplicative functions on almost all short intervals, including Möbius cancellation on almost all intervals whose length tends to infinity arbitrarily slowly [2]. A physical affine fiber has a prescribed origin, and some literal fibers have $O(1)$ length. Neither feature is selected by an almost-all-origin theorem. Moreover, the cited result concerns the mean of one multiplicative function; it is not itself a theorem for the two-sign affine product in (30).

6.3 Principal quantifier mismatches

Input	Native quantifiers	Missing literal certificate
Tao 2016	Fixed nonproportional forms; $1/n$ weight; growing window ratio	Fixed-ratio $o(1)$ logarithmic increments, growing coefficients, actual masks, outer rate
MRT 2015	Ordinary correlations averaged over $h_1, \dots, h_k$	Selection of the prescribed h_0 , growing two-slope forms, actual masks and weights, and outer reassembly
TT 2019	Ordinary correlations outside a zero-log-density set of scales	Every prescribed physical scale, growing forms, literal masks and weights, outer reassembly, and a uniform fixed-power rate
MR 2016	One multiplicative function on almost all short-interval origins	The two-sign affine product, prescribed origin, growing forms, literal masks and weights, outer reassembly, and $O(1)$ -length fibers

Table 1: A theorem can be strong in its native quantifiers without controlling the literal signed Cesàro defect.

Proposition 6.1 (Averages do not select a literal member). *An estimate*

$$\frac{1}{H} \sum_{h \leq H} |B_X(h)| = o(1)$$

does not logically imply $B_X(h_0) = o(1)$ for one prescribed h_0 . Likewise, a theorem outside a zero-logarithmic-density set of scales does not logically imply its conclusion along a prescribed sparse scale sequence.

Proof. For the first statement set $B_X(h_0) = 1$ and $B_X(h) = 0$ otherwise. For the second, put the prescribed sparse sequence into the exceptional set. For example, $\{2^k : k \geq 0\}$ has zero logarithmic density because

$$\frac{1}{\log X} \sum_{2^k \leq X} \frac{1}{2^k} \rightarrow 0.$$

□

Remark 6.2 (Logical versus arithmetic countermodels). [Proposition 6.1](#) classifies a quantifier gap. It does not assert that an actual Möbius correlation is exceptional at h_0 or at the selected physical scales.

7 The literal growing fixed- h_0 interface

The exact Tauberian theorem is useful only after the physical coefficient has been retained. We therefore formulate the crosswalk without claiming that its arithmetic hypothesis is known.

7.1 The affine fibers

After one physical column is opened, the literal TPC-33 affine correlation has the schematic form

$$\mathfrak{a}_{\theta,X}(r) = \mu(\tilde{d}_{\theta,X} + s_{\theta,X}r) \mu(\tilde{u}_{\theta,X} + \alpha_{\theta,X}r) M_{\theta,X}(r) W_{\theta,X}(r), \quad (30)$$

where $M_{\theta,X}$ contains the actual coprimality, residue, and joint masks, $W_{\theta,X}$ contains the physical smooth and logarithmic weights, and

$$s_{\theta,X} \tilde{u}_{\theta,X} - \alpha_{\theta,X} \tilde{d}_{\theta,X} = h_0 \neq 0. \quad (31)$$

The parameter θ also records the opposite row, polarization, divisor layer, interval component, and native keys. The slopes and intercepts may grow with X .

For an ordered fiber of length $N_{\theta,X}$, absorb every literal local factor into a sequence $b_{\theta,X}(n)$, $1 \leq n \leq N_{\theta,X}$, and define

$$S_{\theta,X}(N) = \sum_{n \leq N} b_{\theta,X}(n), \quad L_{\theta,X}(N) = \sum_{n \leq N} \frac{b_{\theta,X}(n)}{n}. \quad (32)$$

This definition does not assert that an existing logarithmic correlation theorem applies to $b_{\theta,X}$.

7.2 An exact conditional certificate

Theorem 7.1 (Literal Tauberian reassembly certificate). *Suppose the complete raw hard zero mode has an exact reassembly*

$$Z_X = \sum_{\theta \in \Theta_X} c_{\theta,X} S_{\theta,X}(N_{\theta,X}) + E_{\text{content},X}, \quad (33)$$

with all native keys and both polarizations retained. Then

$$|Z_X| \leq \sum_{\theta \in \Theta_X} |c_{\theta,X}| N_{\theta,X} \left| \mathcal{T}_{N_{\theta,X}}(L_{\theta,X}) \right| + |E_{\text{content},X}|. \quad (34)$$

The signed defect in (34) may be replaced by the absolute oscillation $\mathcal{A}_N$, the right side of (15), or the right side of (19).

Proof. By [theorem 3.2](#),

$$S_{\theta,X}(N_{\theta,X}) = N_{\theta,X} \mathcal{T}_{N_{\theta,X}}(L_{\theta,X}).$$

Insert this identity into (33), then take one triangle inequality after the complete reassembly. The alternative bounds follow from [corollary 3.3](#) and [theorems 4.3](#) and [4.7](#). □

Remark 7.2 (What is exact and what is conditional). The passage from (33) to (34) is finite L0 algebra. The claim that (33) is the actual raw physical packet, with no missing mask or renormalization, is an L1 verification. A uniform power bound for the defects of the growing Möbius family (30) is an L2 arithmetic estimate and is not proved here.

Remark 7.3 (Short fibers). The asymptotic condition $\omega_N(L) \rightarrow 0$ says nothing about a fiber whose length stays $O(1)$. Such fibers must be returned by an exact finite estimate or controlled after a separate outer reassembly. Their mere cardinality is not a cancellation theorem.

Remark 7.4 (Weights and origins). The harmonic factor $1/n$ in (32) is tied to the chosen local ordering. Partial summation can transfer some smooth weights only when the required prefix estimates are uniform. It cannot silently change the prescribed interval origin, delete the joint mask, or freeze coefficients that grow with X . In particular, the $1/n$ in a fixed-form logarithmic theorem is attached to the original arithmetic variable. After translating a physical fiber with origin $r_{\theta,X}$, the inherited weight is $1/(r_{\theta,X} + n)$, not $1/n$. Replacing one by the other requires an exact change of variables or a uniform partial-summation lemma for the same prefixes; it is not a formal relabeling.

7.3 Fixed-power and endpoint ledger

The exact identity itself spends no fixed power. A useful physical application, however, must prove a rate that survives the full outer sum. For example, if

$$\omega_M(L_{\theta,X}) \ll M^{-\eta+o(1)}, \quad 0 < \eta < 1,$$

uniformly in every literal long fiber and every relevant prefix $2 \leq M \leq N_{\theta,X}$, then [corollary 4.4](#) gives the same per-fiber power. Losses from the number and size of fibers, masks, short-fiber treatment, content completion, and calibrated downstream drift remain to be charged separately.

At the critical physical scales

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)},$$

the recorded two-exponent transfer takes the form

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q^2, \quad D_X \geq X^{-\lambda_D+o(1)} Q^3 J^2 \quad (35)$$

and yields

$$\frac{|Z_X|^2}{D_X} \leq X^{1/400+\lambda_D-2\eta_Z+o(1)}. \quad (36)$$

This is a sufficient ledger, not a claim that the displayed exponents are optimal; see TPC-81 and TPC-82 [8, 9]. Retaining the baseline exponent in (36) requires the determinant-flatness compatibility

$$\lambda_D \leq 2\eta_Z. \quad (37)$$

This is one ledger: it compares the certified lower loss for D_X with the certified cancellation gain for Z_X .

Definition 7.5 (Strict physical endpoint rule). Let Λ_{phys} be the sum of all fixed-power losses actually incurred by localization, coefficient growth, masks, short-fiber treatment, outer reassembly, content completion, dictionary transfer, and downstream calibrated drift. A proposed route continues only if

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}}. \quad (38)$$

Equality or excess is a stop condition. This is a second, independent ledger. A loss already included in $\lambda_D - 2\eta_Z$ is not charged a second time; an interface or downstream loss absent from those exponents must be entered in Λ_{phys} .

Proposition 7.6 (Success checklist). *A logarithmic-to-ordinary advance qualifies as a literal L1 or L2 input only if it returns:*

- (i) *the same fixed h_0 and actual growing affine forms;*
- (ii) *the ordinary physical fibers with their prescribed origins;*
- (iii) *all periodic, coprimality, residue, smooth-weight, and joint-mask factors;*
- (iv) *both polarizations and every native outer key;*
- (v) *a separate exact or outer estimate for bounded-length fibers;*
- (vi) *a quantitative signed-defect or stronger local bound uniform over the active family and all prefixes used by the dyadic recursion;*
- (vii) *a complete fixed-power ledger satisfying (38).*

Proof. Items (i)–(v) ensure that the sequence in (32) is the literal coefficient rather than a frozen or averaged substitute. Item (vi) supplies the hypothesis of [theorem 7.1](#); qualitative normalized logarithmic cancellation alone is ruled out by [corollary 5.4](#). Item (vii) is the endpoint rule. $\square$

8 Claim boundary and conclusion

8.1 What is proved

The following statements are unconditional for arbitrary complex sequences:

- (1) the exact discrete Abel identity

$$S_N = NL_N - \sum_{m < N} L_m;$$

- (2) the necessary-and-sufficient signed Cesàro criterion

$$S_N/N \rightarrow 0 \iff L_N - \frac{1}{N} \sum_{m < N} L_m \rightarrow 0;$$

- (3) convergence of L_N is sufficient, with the quantitative bounds (11) and (12);
- (4) terminal-half logarithmic cancellation and vanishing dyadic logarithmic variation are sufficient, with the explicit bounds (15) and (19);
- (5) a fixed-power terminal-half bound below exponent one transfers without a fixed-power loss;
- (6) the bounded dyadic sequence (21) has normalized logarithmic cancellation on every growing multiplicative window but no ordinary cancellation.

These are L0 identities, inequalities, and a structure-free L0 countermodel.

[Theorem 7.1](#) is also an exact finite implication. Its status becomes L1 only after (33) is verified for the complete raw hard physical packet.

8.2 What is not proved

This paper does not prove:

- (a) terminal-half $o(1)$ or fixed-power cancellation for the actual growing affine Möbius family;
- (b) uniformity of Tao's fixed-form theorem in the polynomially growing physical coefficients;
- (c) selection of the prescribed h_0 from an averaged-shift theorem;
- (d) removal of the exceptional scales or interval origins in an almost-all theorem;
- (e) cancellation on the $O(1)$ -length physical fibers;

- (f) a complete physical outer-reassembly bound;
- (g) a determinant-energy lower bound for D_X ;
- (h) a fixed- h_0 Hardy–Littlewood asymptotic;
- (i) a breach of the general sieve parity barrier;
- (j) a prime-pair lower bound or the twin-prime conjecture.

No conclusion is specialized from a general fixed nonzero h_0 to $h_0 = 2$.

8.3 Conclusion

The logarithmic-to-ordinary gap is not an informal change of weight. For the same ordered coefficient sequence it is measured exactly by

$$\mathcal{T}_N(L) = L_N - \frac{1}{N} \sum_{m < N} L_m.$$

Global boundedness of L_N , or even normalized cancellation on every terminal window whose ratio tends to infinity, does not force this defect to vanish. The smallest exact target is therefore the signed defect itself. Terminal-half cancellation and dyadic variation are stronger, quantitative certificates that can be checked without ambiguity.

For the TPC route this produces a clean fork. A positive advance must prove one of these certificates uniformly for the literal growing fixed- h_0 affine fibers and survive the full outer reassembly. A negative advance may prove that a proposed logarithmic input cannot meet that certificate under its native quantifiers. Either outcome must retain the actual coefficient, global normalization, and the strict endpoint rule $\Lambda_{\text{phys}} < 1/400$. The present paper identifies that gate; it does not cross the unresolved arithmetic parity boundary.

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